

Automask: Promptable Concept Segmentation with SAM3 for Glacier 3D Reconstruction from UAV Imagery

SiCheng Zhao^{1,2,3,†}, Matthew Byrne^{2,†}, and Soumyabrata Dev^{1,2,3}

¹*The ADAPT Research Centre, Dublin, Ireland*

²*School of Computer Science, University College Dublin, Dublin, Ireland*

³*School of Computer Science and Statistics, Trinity College Dublin, Dublin, Ireland*

[†]*These authors contributed equally to this work.*

Abstract

Photogrammetric surveys are the standard way to measure glacier retreat and surface change, but the sky, survey aircraft, and ground crew that appear in the imagery destabilise Structure-from-Motion (SfM) pose estimation and leave halos and ghosting in the reconstructed radiance field. Masking every frame by hand, the standard remedy, costs an analyst one to two hours per 500-image survey and scales linearly with dataset size. We show that this manual stage can be replaced by a single short text prompt, and quantify what that substitution is worth. *Automask* applies one Promptable Concept Segmentation (PCS) prompt from Segment Anything Model 3 (SAM3) across a whole survey, with no fine-tuning and no per-image intervention, ahead of an unmodified COLMAP and Gaussian Splatting pipeline. On a 589-image uncrewed aerial vehicle (UAV) glacier survey, prompted masking improves every metric over an unmasked baseline across ten end-to-end runs: peak signal-to-noise ratio (PSNR) by ~ 1.2 dB, structural similarity index (SSIM) by 0.065, Learned Perceptual Image Patch Similarity (LPIPS) by 0.096, and mean SfM reprojection error from 0.556 px to 0.506 px. We also characterise the glacier-specific failure modes, chiefly ice read as sky under overcast lighting, that bound what a fully automatic single-prompt run can achieve. The implementation is publicly available at <https://github.com/AppleJack9050/automask>.

1 Introduction

Repeated photogrammetric surveys of glaciers yield metre-scale estimates of front retreat, surface elevation change, and ice-flow velocity, but each survey produces hundreds to thousands of overlapping images that must be reconstructed in 3D [Westoby et al., 2012, Geissler et al., 2021]. The standard pipeline is two-stage: Structure-from-Motion (SfM) [Schönberger and Frahm, 2016] recovers camera poses and a sparse point cloud, after which a dense stage produces a continuous radiance field, increasingly with Gaussian Splatting [Kerbl et al., 2023] rather than Multi-View Stereo [Schönberger et al., 2016]; both stages are typically run through COLMAP [Schönberger and Frahm, 2016, Schönberger et al., 2016].

Glacier imagery stresses both stages. Textureless sky destabilises SIFT correspondences at cloud edges and horizons [Lowe, 2004, Gindraux et al., 2017], inflating bundle-adjustment residuals; moving objects (helicopters, ground crew) introduce location-dependent supervision that survives SfM outlier rejection and reappears as ghosting in the splat optimisation; specular and dynamic surfaces violate photometric consistency; and shaded firn or smooth ice under-fires the SIFT detector, leaving the dense stage poorly anchored. Per-image manual masking, the standard remedy, costs an analyst one to two hours per 500-image survey and scales linearly with dataset size.

Promptable foundation models reshape this trade-off. Earlier pipelines chaining a text-conditioned detector [Liu et al., 2024] with a class-agnostic segmenter are brittle on out-of-distribution imagery, whereas Segment Anything Model 3 (SAM3) [Carion et al., 2025] integrates the text-to-region step via its Promptable Concept

Segmentation (PCS) head, which accepts short noun-phrase prompts such as `sky . vehicles . directly`. With image and video propagation, a single prompt can yield coherent masks across an entire survey. The relevant question is therefore not whether such masks appear plausible, but whether unmodified PCS prompting materially improves downstream 3D reconstruction.

We make three contributions: (i) *Automask*, a containerised semi-automatic masking service combining SAM3 with a queue-driven batch processor and a Vue.js editor, designed to integrate with existing pipelines (camera $\rightarrow$ Automask $\rightarrow$ COLMAP $\rightarrow$ Gaussian Splatting) and scale via Kubernetes/KEDA; (ii) an end-to-end evaluation on a 589-image uncrewed aerial vehicle (UAV) glacier survey, reporting peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), and reprojection error over ten runs; and (iii) an analysis of the residual failure modes that bound the reliability of a fully automatic single-prompt run, chiefly ice misclassified as sky under overcast lighting and thin over-bright structures being missed.

2 Design

2.1 Pipeline Overview

An input dataset $\mathcal{I} = \{I_1, \dots, I_N\}$ flows through three stages: (1) **Masking**, in which a SAM3 PCS prompt p and action $a \in \{\text{REMOVE}, \text{KEEP}, \text{HIGHLIGHT}\}$ yield I'_i ; (2) **SfM**, in which COLMAP recovers cameras and a sparse cloud; (3) **Gaussian Splatting**, in which Gaussians initialised from the sparse cloud are optimised against the masked images. The user interacts only with stage 1; SfM and splatting run unchanged from their public implementations (COLMAP 3.13 and `gsplat`).

2.2 SAM3 Masking

For each prompt p , the PCS head returns binary masks $\{m_{i,k}\}_k$ per matched concept instance. In batch mode, p is fixed and SAM3’s internal features enforce coarse spatial and temporal consistency across views, which is what makes a single-prompt, full-dataset workflow tractable. The action a composes masks into I'_i : REMOVE zeroes masked pixels (used when the prompt names unwanted content); KEEP is its dual (occasionally useful in interactive refinement); HIGHLIGHT alpha-blends a random colour over each region, biasing the SIFT detector toward those regions by raising local image entropy.

2.3 Prompt and Action

The configuration evaluated in this paper is background removal: $p = \text{sky . vehicles .}$ with $a = \text{REMOVE}$. It targets the dominant noise sources identified in Section 1, on the hypothesis that SfM is misled by spurious matches in these regions. The KEEP and HIGHLIGHT actions are available in the system and are used during interactive refinement, but are not evaluated here; this configuration is referred to below as “Masked”.

2.4 Reconstruction Backend

COLMAP uses default SIFT settings, untuned per dataset. The calibrated cameras and sparse cloud initialise Gaussian Splatting (PyTorch, `gsplat`, default optimiser settings). When REMOVE has been applied, masked pixels do not contribute to the photometric loss, so the splat optimisation benefits from cleaner supervision as well as cleaner poses.

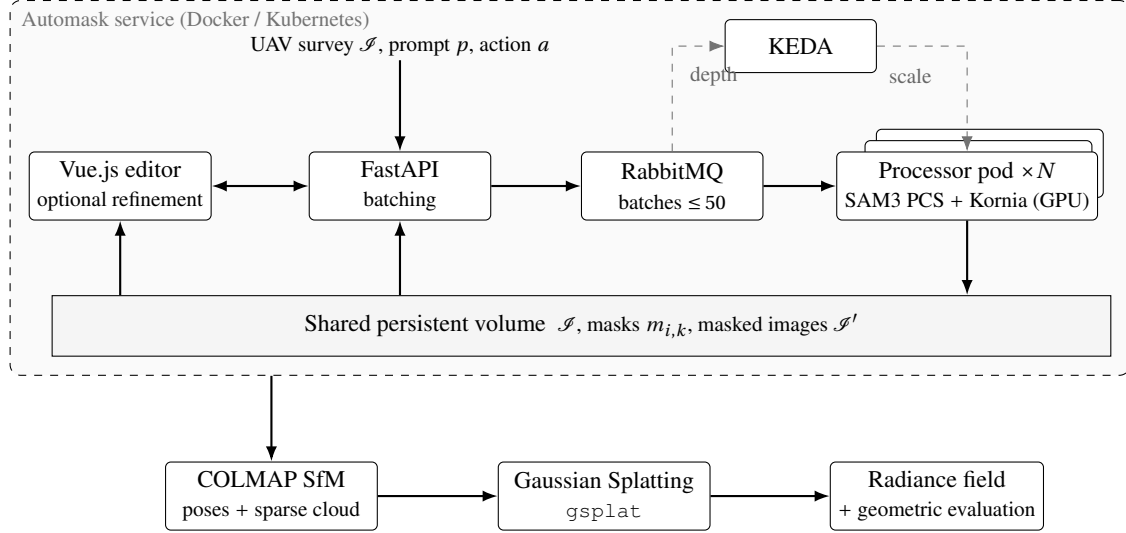


Figure 1: Automask architecture. Solid arrows carry image data, dashed arrows carry control signals. One prompt and action pair is applied to a whole survey: the API chunks the dataset into batches, RabbitMQ distributes them across processor pods, and KEDA adds pods when the backlog exceeds five batches. Masked images are written to the shared volume and consumed unmodified by the reconstruction backend; the editor is optional and operates on individual images.

2.5 Evaluation Metrics

We use PSNR (pixel fidelity, dB), SSIM (local luminance/contrast/structure), LPIPS (AlexNet-backbone perceptual similarity, sensitive to blur and ringing that PSNR/SSIM miss), and SfM reprojection error:

$$\text{RMSE} = \frac{1}{|\mathcal{O}|} \sum_{(i,j) \in \mathcal{O}} \|x_{ij} - \bar{x}_{ij}\|, \quad (1)$$

where $\mathcal{O}$ is the set of valid observations, x_{ij} the observed position of point i in image j , and $\bar{x}_{ij}$ its prediction.

3 Implementation

Automask separates three concerns (a low-latency interactive front-end, a stateful API, and a GPU-backed image processor) into Docker services orchestrated on Kubernetes, communicating through a RabbitMQ queue and a shared persistent volume (Figure 1). Only the masking processor is specific to the vision task; the remaining components exist so that one prompt can be applied to an entire survey without analyst involvement, which is the property the evaluation in Section 4 depends on.

3.1 Masking Processor

Each processor pod runs a consumer that hands batches to a `FileProcessor`, which processes files serially within the pod; parallelism is provided across pods rather than within them. An empty prompt invokes SAM3’s automatic mask generator; a non-empty prompt goes through PCS directly. Masks are composed with the action by a `FileEditor` built on Kornia [Riba et al., 2019], keeping the pipeline on the GPU and avoiding a host round trip per image. The three actions reduce to tensor operations: `HIGHLIGHT` alpha-blends a per-mask random colour at $\alpha = 0.5$; `KEEP` multiplies by the mask union; `REMOVE` multiplies by its complement. Per-image cost is dominated by SAM3 inference, so the processor is compute-bound and scales close to linearly in the number of pods.

3.2 Batch Execution and Scaling

The API chunks each request into batches of up to 50 files. Batching bounds per-message GPU memory, lets multiple pods work in parallel on one user’s request, and gives RabbitMQ enough granularity to redistribute work after pod failure. KEDA monitors queue depth and adds processor pods when the backlog exceeds five queued batches (~ 250 images). A persistent volume mounted in every pod holds input, intermediate, and output images, so newly scheduled pods resume without per-pod data migration. Processors are stateless except for the GPU-resident SAM3 checkpoint, which is baked into the image at build time to keep cold start deterministic; horizontal scaling therefore requires no coordination between pods.

3.3 Interactive Refinement

The Vue.js editor exposes four file states (Uploaded, Processing, Processed, Saved). From Processed, the user opens an editor displaying the source image, the SAM3 masks, and refinement tools; per-mask remove and keep operations execute client-side, and a brush tool handles cases the prompt misses, such as the thin structures discussed in Section 5. Edits are recorded in an undo/redo store that persists across page navigation. The editor is used for per-image correction only and plays no part in the batch results reported below, which use a single unedited prompt.

4 Evaluation

4.1 Setup

We use a 589-image UAV glacier survey (DJI FC330, pinhole with two radial distortion coefficients) from the UNAVCO TLS repository, downsampled to 2000×1500 pixels. Approximately one-third of frames include sky, a smaller fraction the survey helicopter, and the glacier surface ranges from crevassed sections to smooth shaded firn. All experiments run on a single workstation (NVIDIA RTX 5090, AMD Ryzen 7 9700X, 48 GB RAM, Ubuntu 24.04.3) using COLMAP 3.13 (default SIFT), `gsplat` on PyTorch 2.8/CUDA 12.9 (default optimiser), and the public SAM3 checkpoint with no fine-tuning.

Both configurations are run through the full pipeline ten times to control for splat stochasticity. PSNR, SSIM, and LPIPS are computed on the original unmasked held-out views, so both configurations are scored over an identical pixel set; masking affects training supervision only.

4.2 Full-Pipeline Quantitative Results

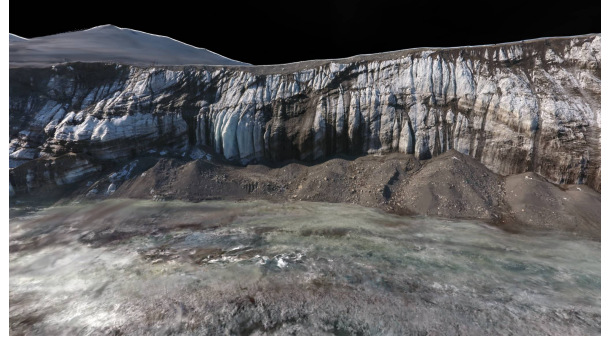
Table 1: Reconstruction quality on the 589-image UAV glacier dataset. Mean $\pm$ standard deviation over ten independent runs; best in bold.

Dataset	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Reproj. $\downarrow$
Original	22.30 ± 0.18	0.723 ± 0.006	0.408 ± 0.014	0.556 ± 0.012
Masked	23.52 ± 0.21	0.788 ± 0.004	0.312 ± 0.015	0.506 ± 0.010

Masking improves every metric (Table 1): PSNR by ~ 1.2 dB, SSIM by 0.065, LPIPS by 0.096, reprojection error from 0.556 to 0.506 px. Masked variance is comparable to or smaller than Original’s on every metric, so the gain is not driven by a few favourable runs. Expressed as standardised effect sizes, the four improvements span $d \approx 4.5$ (reprojection error) to $d \approx 12.7$ (SSIM): modest in absolute terms, but large relative to the run-to-run variation of the splat optimisation. SSIM shows the smallest variance (consistent with its smoothness as a metric) and LPIPS the largest (deep-feature metrics are more sensitive to optimiser-trajectory changes).



(a) Unmasked reconstruction.



(b) Masked reconstruction.

Figure 2: Renderings from comparable viewpoints. Unmasked (left) shows blurred halos and ghosting from sky pixels and dynamic objects entering the splat optimisation; these are absent on the right.

4.3 Qualitative Results

Figure 2 shows two patterns the per-pixel metrics only partially capture: surviving sky pixels produce diffuse blue halos along the glacier silhouette (low-opacity Gaussians instantiated at indeterminate depth), and dynamic content produces ghosts where the same 3D location receives inconsistent supervision across frames. Both are largely absent from the masked reconstruction.

5 Discussion

Magnitude of the gains. The improvements are consistent but modest in absolute terms. Viewpoint diversity and inter-image overlap dominate reconstruction quality, and preprocessing cannot manufacture them; masking contributes at the margin by removing avoidable noise. What remains is bounded by the failure modes below, which suggests that further gains lie in improving model behaviour where it fails rather than in further prompt engineering.

Failure modes. Two glacier-specific failures recur. *Ice misclassified as sky*: foreground ice is occasionally classified as sky under overcast lighting, where the statistics of grey cloud overlap with shaded firn or wet ice; PCS does not distinguish visually similar concepts when one is absent from its training distribution. This motivates domain-specific fine-tuning of the kind exemplified by MedSAM [Ma et al., 2024]. *Thin over-bright structures missed*: helicopter rotors, guide wires, and antenna edges are at the segmentation model’s resolution threshold and are consistently missed by the `vehicles` prompt even when the helicopter body is segmented. Both are easily corrected with a brush stroke in the editor but constrain what a fully automatic single-prompt run can achieve.

Limitations and future work. The evaluation uses a single dataset and a single prompt and action configuration; alternative prompts, alternative actions, and their composition are not evaluated here, and the result may not transfer to substantially different content distributions. SfM and splat backends use default settings. The evaluation measures novel-view rendering quality and SfM residuals; the geometric accuracy of the reconstructed surface is not assessed against an independent reference. The most direct extension is glacier-specific fine-tuning of the PCS head, in the manner of MedSAM3 [Liu et al., 2025], for which a curated corpus with hand-verified ice/sky boundaries should suffice. Beyond preprocessing, the SAM3D Objects family suggests an end-to-end pipeline that segments and produces 3D structure with one model, subsuming COLMAP. A natural front-end extension is collaborative editing via a WebSocket that broadcasts edit events through the existing undo/redo store.

6 Related Work

Promptable segmentation. Segment Anything (SAM) [Kirillov et al., 2023] introduced promptable image segmentation as a foundation-model task, separating a heavy ViT image encoder (run once per image) from a lightweight prompt encoder and mask decoder so image embeddings can be reused across prompts. SAM3 [Carion et al., 2025] shifts the centre of gravity to Promptable Concept Segmentation, which accepts noun-phrase prompts and/or image exemplars and returns one mask per matching instance, using a multimodal LLM [Yin et al., 2024] for ambiguous prompts and a DETR-style detector [Carion et al., 2020] with a learned “presence token”. The PCS head subsumes the role of a separate open-vocabulary detector, the property Automask exploits to drop a stage relative to a chained text-detector + segmenter design.

Domain-adapted SAM variants. General-purpose SAM checkpoints fail on data far from natural images. MedSAM [Ma et al., 2024] adapts SAM to medical imaging via encoder downsampling, an FPN for multi-scale lesion segmentation, and CT/MRI fine-tuning. SAM-Adapter [Chen et al., 2023] freezes foundation weights and injects task-specific knowledge through small MLP adapters. We considered SAM-Adapter for ice/water boundaries but found no clean integration with PCS-style prompting; the results in Section 4 suggest fine-tuning is the appropriate next step.

Applications and SfM preprocessing. Existing SAM-based applications cluster into browser-based single-image demos (often using SlimSAM [Chen et al., 2024] or ONNX runtime, unsuitable for batch processing hundreds of images) and vertical clinical tools such as MedSAM-Lite. Automask occupies the middle: batch input with per-image interactive refinement for remote-sensing researchers. SfM preprocessing pipelines typically use fixed-class semantic networks or heuristic colour models for sky removal, both of which generalise poorly to glacial imagery in which overcast sky is hard to separate from snow. Foundation models accept the problem class as a prompt at inference time; the contribution here is therefore not a new architecture but a controlled empirical study of what such a model contributes inside an existing SfM + Gaussian Splatting pipeline.

Glacier remote sensing. Photogrammetric monitoring of glaciers has expanded with UAV adoption [Bhardwaj et al., 2016, Zhao et al., , Zhao et al., 2025]: consumer-grade platforms enable repeated surveys at lower cost than crewed aircraft or terrestrial laser scanning while resolving sub-metre features. Such imagery is dense, high-overlap, and rich in precisely the non-glacier content this work targets.

7 Conclusion

Automask uses SAM3’s PCS head to convert a single short text prompt into per-image masks across an entire glacier survey, improving downstream COLMAP + Gaussian Splatting reconstruction on all four metrics simultaneously across ten runs. The magnitude of the gains is constrained by source-survey viewpoint coverage and by glacier-specific failure modes the unmodified PCS head inherits from its training distribution, which makes fine-tuning rather than further prompt engineering the natural next question. The SAM3 checkpoint is the only stateful component of the image processor, so replacing it is a single configuration change.

Acknowledgments

This research was conducted with the financial support of Research Ireland under Grant Agreement No. 13/RC/2106_P2 at the ADAPT Centre at Trinity College Dublin. ADAPT, the Research Ireland Centre for AI-Driven Digital Content Technology, is funded through the Research Ireland Centres Programme. For the purpose of Open Access, the author has applied a CC BY public copyright licence to any Author Accepted Manuscript version arising from this submission. This work is partly supported by China Scholarship Council (CSC) (Grant No. 202408320050).

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